		Phase difference $= 2\pi \times \frac{\Delta x}{\lambda} = 2\pi \frac{0.75\lambda}{\lambda} = 1.5\pi$
14	**C**	$$\frac{A_2}{A_1} = \left(\frac{L_2}{L_1}\right)^2 = \left(\frac{16}{2}\right)^2 = 64$$ $$I = \frac{P}{A} \Rightarrow I \propto \frac{1}{A} \Rightarrow \frac{I_2}{I_1} = \frac{A_1}{A_2} = \frac{1}{64} \Rightarrow I_2 = \frac{I}{64}$$ $$I \propto x_0^2 \Rightarrow \frac{I_2}{I_1} = \left(\frac{x_2}{x_1}\right)^2 \Rightarrow \frac{x_2}{x_1} = \sqrt{\frac{I_2}{I_1}} = \sqrt{\frac{1}{64}} = \frac{1}{8} \Rightarrow x_2 = \frac{A}{8}$$
15	**C**	Wavelength of closed pipe is $\lambda = 4L$, so speed of sound $v = f\lambda = 4fL$.